CHAPTER

30

Complex Numbers

Complex numbers are an extension of the ordinary numbers used in everyday math. They have the unique property of representing and manipulating *two* variables as a *single* quantity. This fits very naturally with Fourier analysis, where the frequency domain is composed of two signals, the real and the imaginary parts. Complex numbers shorten the equations used in DSP, and enable techniques that are difficult or impossible with real numbers alone. For instance, the Fast Fourier Transform is based on complex numbers. Unfortunately, complex techniques are very mathematical, and it requires a great deal of study and practice to use them effectively. Many scientists and engineers regard complex techniques as the dividing line between DSP as a *tool*, and DSP as a *career*. In this chapter, we look at the mathematics of complex numbers, and elementary ways of using them in science and engineering. The following three chapters discuss important techniques based on complex numbers: the *complex Fourier transform*, the *Laplace transform*, and the *z-transform*. These complex transforms are the heart of theoretical DSP. Get ready, here comes the math!

The Complex Number System

To illustrate complex numbers, consider a child throwing a ball into the air. For example, assume that the ball is thrown straight up, with an initial velocity of 9.8 meters per second. One second after it leaves the child's hand, the ball has reached a height of 4.9 meters, and the acceleration of gravity (9.8 meters per second^2) has reduced its velocity to zero. The ball then accelerates toward the ground, being caught by the child two seconds after it was thrown. From basic physics equations, the height of the ball at any instant of time is given by:

$$h = \frac{-gt^2}{2} + vt$$